

Revised Project Plan

Look-alike Domain Analyzer & Reporting Tool

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1. Project Overview

This document is the revised project plan for Phase 2 of our Lookalike Domain Analyzer and Reporting Tool. It contains a review of Phase 1 deliverables, instructor feedback, and an updated scope and timeline for the current semester.

1.1 What We Are Building

The project is a Chrome extension that allows users to analyze domains and URLs for lookalikes and phishing risks. The tool serves two audiences:

- End users: User submits a domain or URL through the extension and receives an easy-to-understand report that explains if the domain is suspicious and should be avoided.
- Security analysts: Same submission provides a full evidence breakdown covering threat intelligence, heuristic signals, infrastructure data, WHOIS and registrar information, MX records, blacklist status, and a complete risk map suitable for investigations and reporting.

Both modes are powered by the same backend analysis pipeline, the level of detail shown is tailored to each audience.

1.2 Key Components

- Chrome extension — manual URL/domain entry, brand override, JSON export, and lookalike candidate display
- AWS-hosted TypeScript backend — Lambda handlers, API Gateway, CDK infrastructure stack, deployed in central-1
- Deterministic scoring engine — weighted signals across lexical, infrastructure, content, reputation, behavioral, and history categories
- AWS Bedrock LLM — used for feature collection and analysis of evidence contributing to the threat score, and for report formatting
- Legitimate lookalike generation — DNSTwister-based generation of existing potential lookalikes for a submitted legitimate domain
- Free and open intelligence sources — RDAP, crt.sh, Internet Archive CDX API, OpenPhish, and PhishTank
- Optional paid intelligence providers — VirusTotal, AbuseIPDB, Google Safe Browsing (key-gated, graceful degradation when unavailable)

2. Phase 1 Summary – What Was Completed

Phase 1 produced a working proof of concept that established the foundational architecture and analyst workflow. The following capabilities are not fully optimized but were built and demonstrated at the end of the phase:

- Manual URL/domain analysis
- Brand recognition and override
- Deterministic scoring
- Analyst-facing evidence display
- JSON export
- DNSTwister-based lookalike generation

3. Feedback Received and How It Was Addressed

The following summarizes the key feedback received at the end of Phase 1 and the corresponding adjustments made to Phase 2 scope and approach.

Criticism: Proof of concept demonstration was simple and does not stand out among similar existing tools.

Response: We will continue developing and improving the functionality and design of the extension to ensure all planned features are implemented. The focus of Phase 2 is to build features that existing tools do not combine in one place.

Recommendation: Add a feature that visits the lookalike website to observe behaviour as an addition to the threat score logic.

Response: We will implement a sandbox environment that visits the lookalike website to observe behaviours and evaluate the overall appearance of the site as an input to the threat score.

Recommendation: Alert users with the extension enabled that they may be visiting a potential lookalike or malicious website.

Response: As part of our Chrome extension, there will be a live monitoring capability that alerts the user of a potential lookalike, recommending they leave the website or providing a warning based on the severity of the threat score.

4. What Still Needs to Be Completed

Phase 2 closes the following gaps identified from Phase 1 and the feedback review:

4.1 Technical

- Optimized accuracy of threat score logic and fine-tuned severity levels.
- Recognition of brand as the correct existing entity being impersonated.

4.2 Infrastructure

- Finalized AWS environment.

4.3 Design

- A completed Chrome extension with two depth levels: a full analyst view and a simplified end-user view.
- A comprehensive dashboard showcasing the tool's full capability, featuring granular threat score analysis and a reporting tool for security analysts.

4.4 Reporting

- A functional reporting tool that exports evidence and information into universal JSON format for time-effective takedown procedures.

5. Final Deliverables

The following artifacts will be produced by the end of Phase 2:

Deliverable	Description
Chrome Extension	Chrome extension built against the deployed production API, with both the basic user view and the full analyst view implemented, and the reporting module integrated.
Deployed AWS Backend	Live deployment across dev, test, and prod environments with Secrets Manager, CloudWatch, and DynamoDB lifecycle configured.
Basic User Result View	Simplified, plain-language safety result for non-technical users explaining why a domain is flagged and what action to take.
Full Analyst Result View	Complete evidence breakdown including all threat intelligence, heuristic signals, infrastructure data, WHOIS/registrar info, MX records, and blacklist status.
Human-Readable Analyst Report	Exportable analyst report (structured format) in addition to the existing JSON export, suitable for distribution to registrars, brands, or security teams.
Scoring Corpus and Calibration	Labeled sample set used to calibrate and validate scoring weights, documented for reproducibility.
Expanded Test Suite	End-to-end analyst workflow tests and degraded-provider condition tests in addition to existing unit and integration coverage.
Release Documentation	Documented deployment and release process covering packaging, environment promotion, and smoke-test procedures.
Final Demo and Presentation	Live demonstration covering the full Phase 2 workflow: domain submission, basic user view, analyst view, lookalike generation, and report export.

6. How We Plan to Complete the Work

We are organizing Phase 2 into four workstreams across four milestones throughout the allocated thirteen weeks.

6.1 Workstreams

Workstream	Scope
Technical	Provider hardening, signal completion, redirect-chain analysis, content analysis improvements, phishing-template detection, brand catalog expansion, scoring calibration, test coverage expansion.
Infrastructure	AWS environment formalization, Secrets Manager migration, CDK cleanup, DynamoDB lifecycle, CloudWatch deployment, release process documentation, extension packaging against live API.
Design & UX	Basic user simplified view, automatic tab detection, evidence hierarchy redesign, degraded-state UI improvements, lookalike batch review, layout and typography consistency.

Reporting	Registrar and brand contact expansion, CERT/CSIRT routing, Bedrock dependency reduction for core fields, human-readable report format, audience-specific report views.
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6.2 Milestones

Milestone	Focus	Outcome
Milestone 1 (Weeks 1–3)	Provider hardening, infrastructure signal completion, passive history expansion, provider diagnostics cleanup.	Reliable backend evidence model — all signals backed by real logic or explicit failure states.
Milestone 2 (Weeks 4–6)	Content analysis improvements, phishing-template detection, brand-matching expansion, score calibration.	Higher-quality detection and defensible, calibrated scoring.
Milestone 3 (Weeks 7–9)	Basic user view, analyst UX cleanup, lookalike-review improvements, structured report output, expanded reporting coverage.	Both user-facing views complete and a full reporting package.
Milestone 4 (Weeks 10–13)	AWS environment maturity, Secrets Manager, CloudWatch, release packaging, expanded tests, deployment validation, final demo.	Stable Phase 2 release candidate and submission-ready deliverables.

7. Weekly Milestone Breakdown

Each week lists the primary workstream focus and specific tasks assigned to each team member. Tasks map directly to the milestone outcomes above.

Week	Workstream(s)	Jamal Meic (Infrastructure)	Nathanael Churcher (Technical)	Damian Ogorek (Design & UX)	Lily Fulcher (Reporting)
Milestone 1 (Weeks 1–3) — Foundation and Backend Reliability					
Week 1	Technical	Review AWS PoC infrastructure, document gaps in environment setup and Lambda configuration.	Review PoC scoring engine, identify provider gaps, timeout issues, and missing signal coverage.	Audit existing extension UX, document layout problems and missing flows for Phase 2.	Review Phase 1 report output, identify gaps in structure, contact fields, and export format.
Week 2	Technical	Configure API Gateway and Lambda environment for finalized provider set, establish dev deployment.	Complete VirusTotal, AbuseIPDB, and Google Safe Browsing integrations, implement redirect-chain analysis and final-destination resolution.	Produce updated wireframes for basic user view and analyst view based on Phase 1 audit findings.	Research registrar and brand contact sources, draft initial report field schema and routing logic.
Week 3	Technical + Infrastructure	Begin CDK cleanup, confirm backend infrastructure pipeline is stable for ongoing development.	Improve passive DNS and historical ownership intelligence feeding into score logic, confirm backend evidence pipeline.	Refine wireframes based on team feedback, lock layouts for both views ahead of build phase.	Define CERT/CSIRT fallback routing rules, document report format schema for implementation.

Milestone 2 (Weeks 4–6) — Detection and Scoring					
Week 4	Technical	Monitor Lambda performance under content analysis load, support infrastructure for new signal sources.	Improve content analysis: text similarity scoring, asset reuse detection, and form-action classification.	Begin building the basic user view layout in the Chrome extension based on locked wireframes.	Build initial registrar contact catalog data structure, integrate with the report schema defined in Week 3.
Week 5	Technical	Expand brand catalog infrastructure in the backend, validate provider API key configuration and rate-limit handling.	Implement phishing template and visual similarity detection, optimize brand recognition to cover typo and confusable matching.	Continue building basic user view, begin analyst evidence hierarchy redesign and degraded-state UI planning.	Continue registrar contact expansion, research and document CERT/CSIRT routing requirements and fallback conditions.
Week 6	Technical	Support scoring calibration with infrastructure logging, validate signal pipeline accuracy end-to-end.	Finalize detection and scoring accuracy, tune thresholds to cleanly separate real threats from false positives.	Polish both view layouts for consistency, implement initial degraded-state UI for missing provider data.	Draft human-readable report format, finalize CERT/CSIRT routing logic and test against sample contacts.
Milestone 3 (Weeks 7–9) — UX, Reporting, and Environment Planning					
Week 7	Design + Technical	Begin cloud hosting architecture research, support extension-to-backend API integration.	Connect backend scoring output to extension API layer, begin reporting tool backend module.	Improve extension layout with focus on intuitive experience for both basic and analyst audiences.	Begin designing and planning the reporting tool module, map evidence fields to report sections.
Week 8	Design	Continue cloud hosting research, finalize hosting plan and begin Secrets Manager setup planning.	Continue reporting tool backend development, support content analysis improvements and provider reliability.	Improve extension usability: error states, loading indicators, and layout polish for both views.	Integrate registrar contacts into report output, continue developing reporting module with audience-specific sections.
Week 9	Reporting + Design	Finalize cloud hosting implementation plan, hand off architecture decisions to Week 10 build.	Complete backend integration for the reporting module, verify scoring data feeds correctly into report fields.	Finalize both view layouts, begin typography and spacing consistency pass across the full extension.	Finalize reporting module implementation, add CERT/CSIRT contacts and proper routing, optimize contact lookup coverage.
Milestone 4 (Weeks 10–13) — Final Optimization and Delivery					
Week 10	Infrastructure	Implement Secrets Manager for all API keys, clean remaining CDK deprecation warnings, begin deploying finalized cloud hosting.	Validate all provider integrations against Secrets Manager key retrieval, confirm graceful-degradation paths work under the new config.	Begin final extension polish pass, connect extension to the live deployed API for end-to-end testing.	Review report output against the finalized backend, begin generating draft human-readable report samples.
Week 11	Infrastructure + Technical	Configure DynamoDB lifecycle/TTL policies and caching layer, deploy CloudWatch dashboard, log groups, and initial alert set.	Resolve redirect-chain edge cases, harden VirusTotal and AbuseIPDB retry paths, validate caching layer against the scoring pipeline.	Final layout and typography consistency pass across basic and analyst views, tighten degraded-state UI so missing provider data is shown clearly.	Finalize human-readable analyst report format, expand registrar and brand contact catalog, validate CERT/CSIRT routing fallback.
Week 12	Infrastructure + Technical	Run full deployment validation across dev, test, and prod, finalize	Run end-to-end scoring regression against the calibration corpus, fix	End-to-end UX walk-through against the deployed prod API,	Generate sample distribution-ready reports for the demo

		release packaging and smoke-test script, commit release documentation.	remaining false-positive and false-negative patterns.	record basic user and analyst flows for demo materials.	set, verify CERT/CSIRT routing against test contacts.
Week 13	All Workstreams	Final deployment checks, review and submit all release documentation, support demo preparation.	Final scoring verification pass, confirm backend stability for live demo, fix any last-minute regressions.	Finalize demo materials (screenshots, screen recordings), prepare and rehearse presentation slides.	Finalize all report samples, assist with demo script and final submission packaging.

8. Risks, Challenges, and Blockers

Risk	Likelihood	Impact	Mitigation
External provider API rate limits or key access delays	Medium	High	Obtain and validate all API keys by end of Week 2. Implement graceful degradation so the tool remains functional when a provider is unavailable.
LLM inference speed vs. cost trade-off	High	Medium	For Phase 2 the current model tier is acceptable for a capstone demo environment. If latency becomes a blocker, the team will evaluate switching to a more cost-effective hosting platform.